



Powering Innovation That Drives Human Advancement

Get Smart with Automated Tools for SIPI in HFSS 3D Layout

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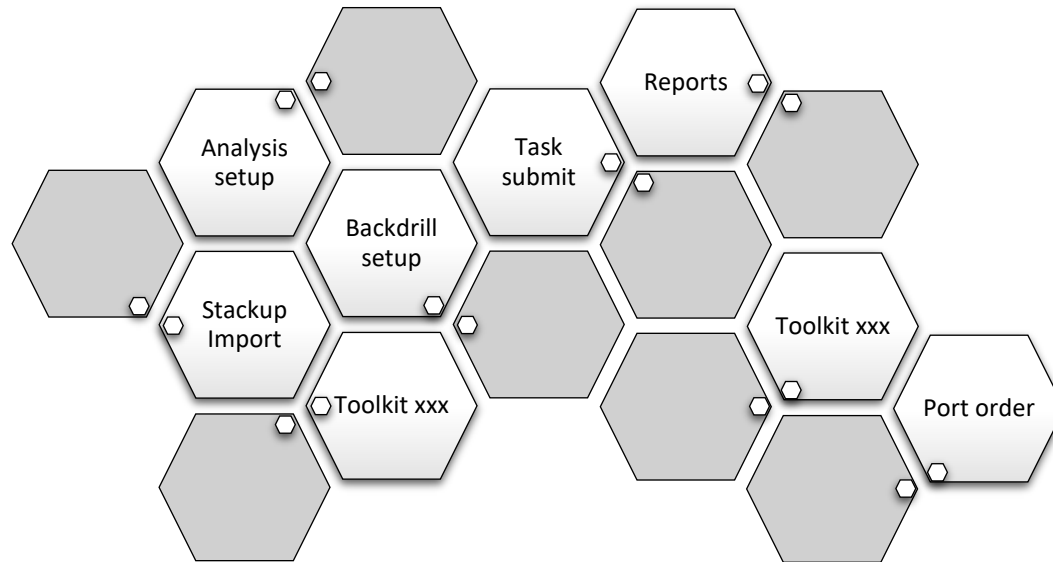
2025-05

Content

- Aedt toolbox Introduction
- Scripts sharing and usage demo
- Manager your own script
- Download

How to run your scripts smartly

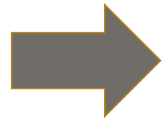
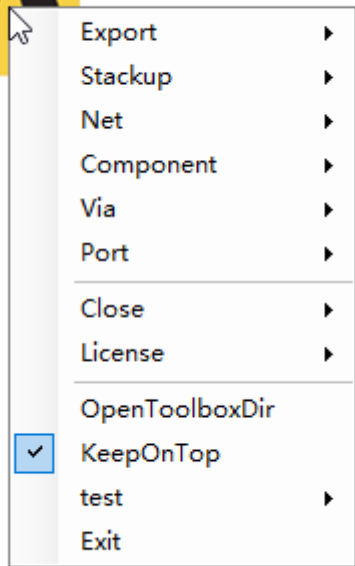
- Automated script files are also a significant asset of the company. Users may be using different versions of scripts, leading to inconsistent results and conclusions, which can cause losses to projects.
 - How to ensure that scripts are distributed to every user?
 - How to update scripts on the user side to ensure synchronization?
 - How to ensure compatibility with different versions of AEDT?



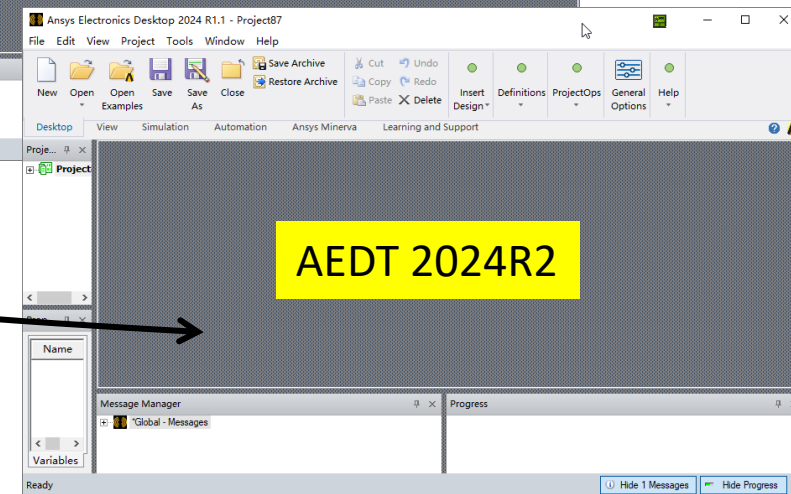
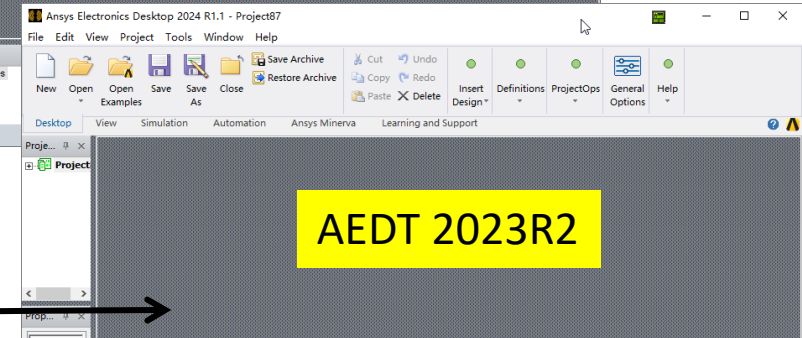
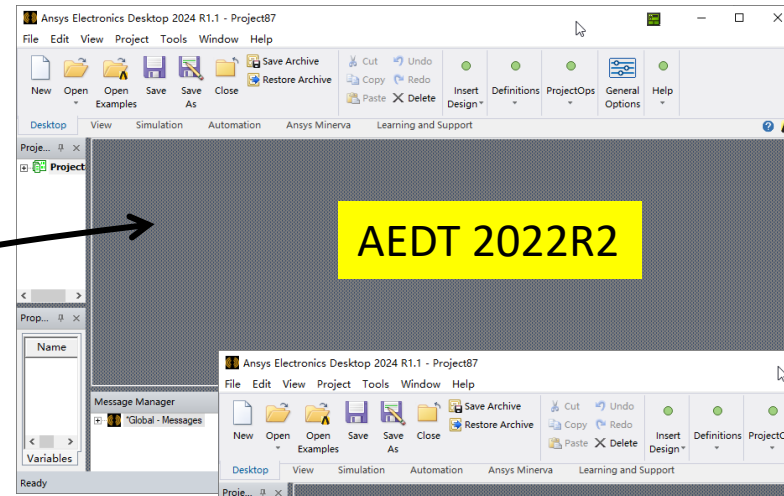
There are many scattered scripts within the company.

Aedt Toolbox

Aedt Toolbox is a tool plugin that provides interfaces for managing and running scripts.



PyAEDT

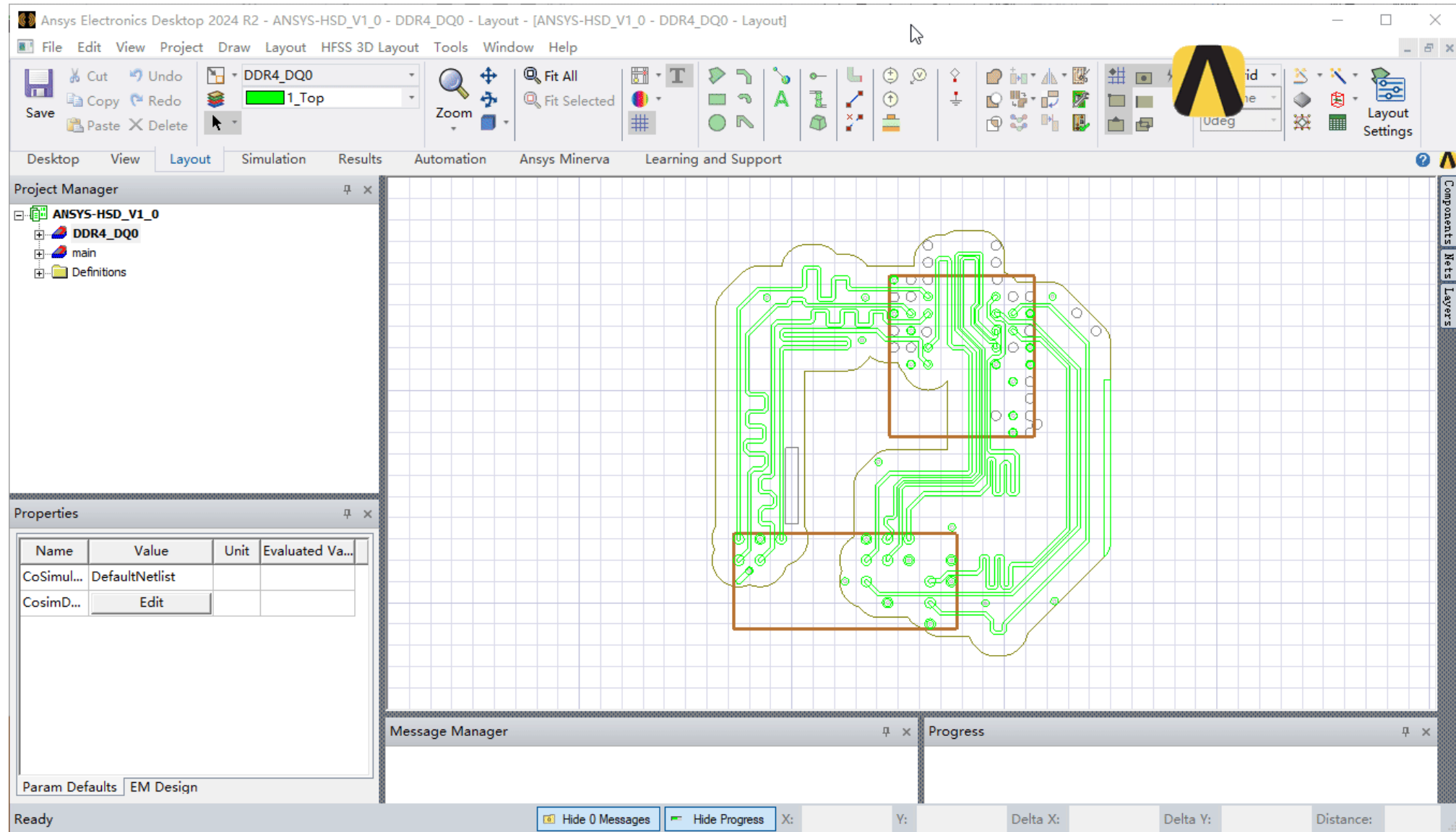


- ✓ Centrally manage scripts to form a common set of tools.
- ✓ Update in real-time to maintain consistency in script usage among users.
- ✓ Ensure compatibility across different AEDT versions..

Scripts in the toolbox now

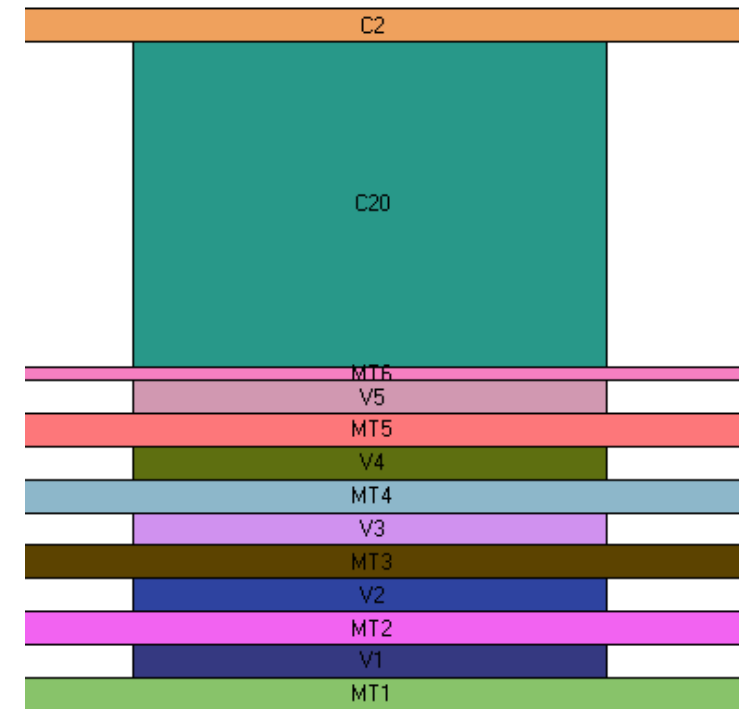
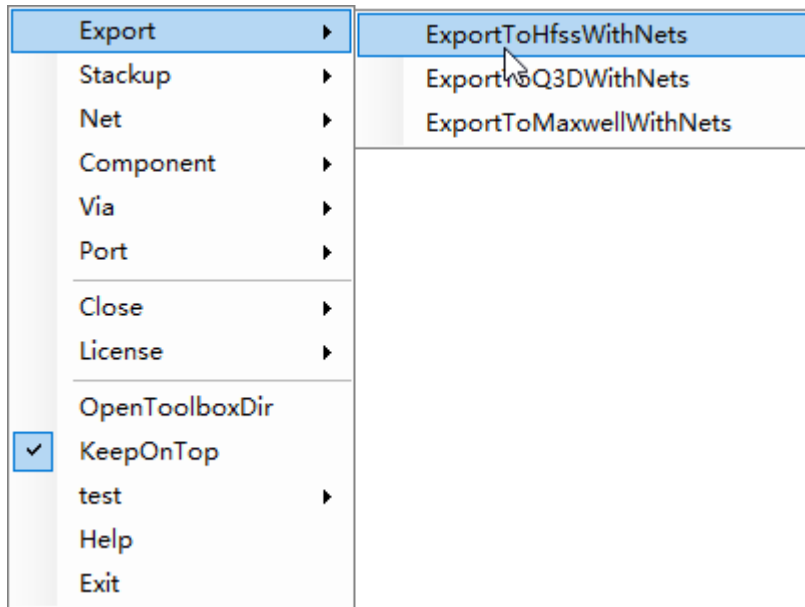
Menu Items	Ansys Tool	Description
ExportToHfssWithNets ExportToQ3DWithNets ExportToMaxwellWithNets	3D Layout	Export to HFSS/Q3D/Maxwell 3D with Nets information
ExportCSVStackup ImportCSVStackup	3D Layout	Import Excel stackup files obtained from factory.
AutoXnet	3D Layout	assist in naming the net after it passes through a capacitor
DissolveInvalidRLC	3D Layout	Removing invalid RLC components only 1 pin
AutoBackdrill	3D Layout	Set up back drilling for vias according to the given Nets(regex) and Stub information.
KeepHDIViaPad	3D Layout	detect HDI vias and add circular shapes on each layer to keep these existing pads.
RenameEdgePort	3D Layout	Rename port name like “Port1” to a meaningful name like “U1.A2.DQ0”
OrderPorts	3D Layout	Sort the Ports in the 3D Layout according to Net for S-parameters

Export 3D structure with Nets information



Export 3D structure with Nets information

- Support importing 3D structures with Nets information from 3D Layout into HFSS, Q3D, and Maxwell. Support both laminated and overlapping stackups.

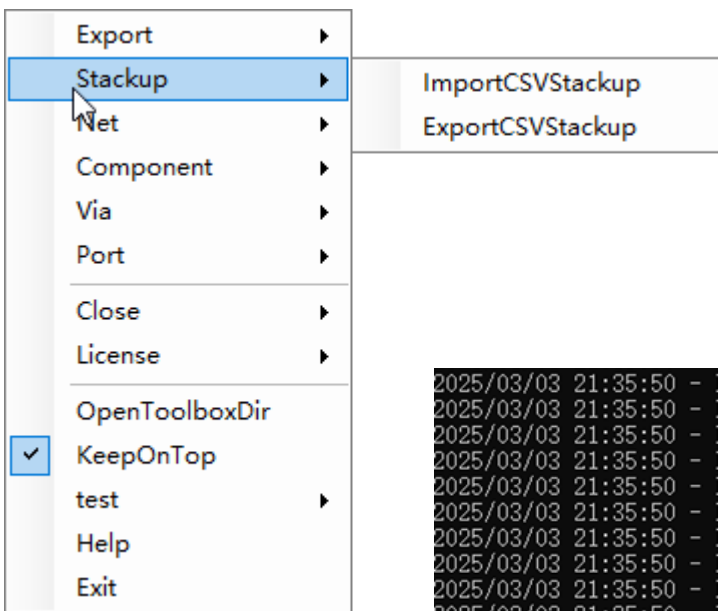


Support both laminated and overlapping stackups

Export/import stackups by Excel

Import Excel stackup files obtained from factory.

1. Use `ExportCSVStackup` to export existing stackup templates.
2. Fill in the existing stackup content according to the format.
3. Use `ImportCSVStackup` to import the modified CSV file.



```
2025/03/03 21:35:50 - INFO: update Conductor layers data by index: L8_SIG
2025/03/03 21:35:50 - INFO: update Conductor layers data by index: L7_GND
2025/03/03 21:35:50 - INFO: update Conductor layers data by index: L6_SIG
2025/03/03 21:35:50 - INFO: update Conductor layers data by index: L5_GND
2025/03/03 21:35:50 - INFO: update Conductor layers data by index: L4_SIG
2025/03/03 21:35:50 - INFO: update Conductor layers data by index: L3_SIG
2025/03/03 21:35:50 - INFO: update Conductor layers data by index: L2_SIG
2025/03/03 21:35:50 - INFO: update Conductor layers data by index: L1_SIG
2025/03/03 21:35:50 - INFO: Recover AutoSave for function: loadFromDict
2025/03/03 21:35:50 - INFO: loadFromDict: Process time 0.869063138961792s
```

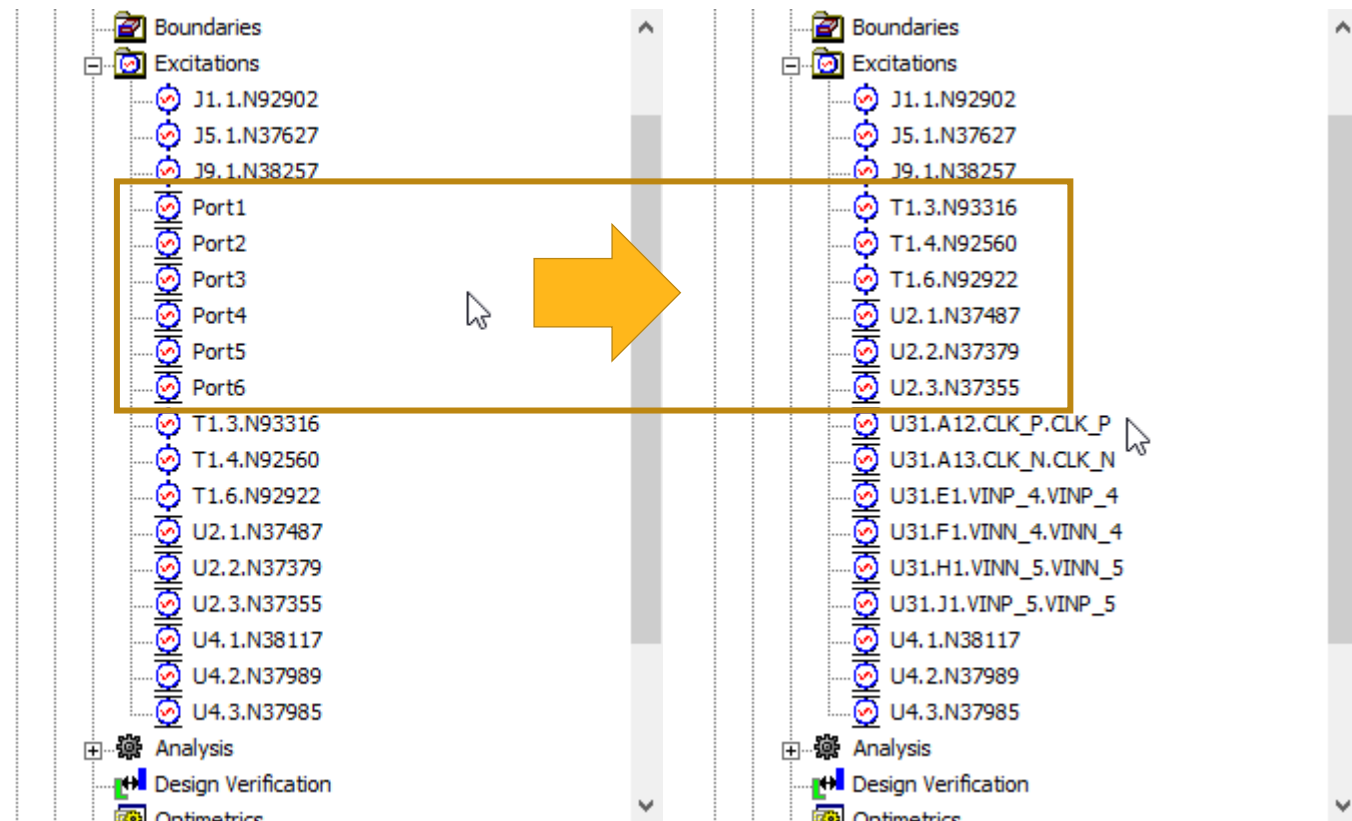
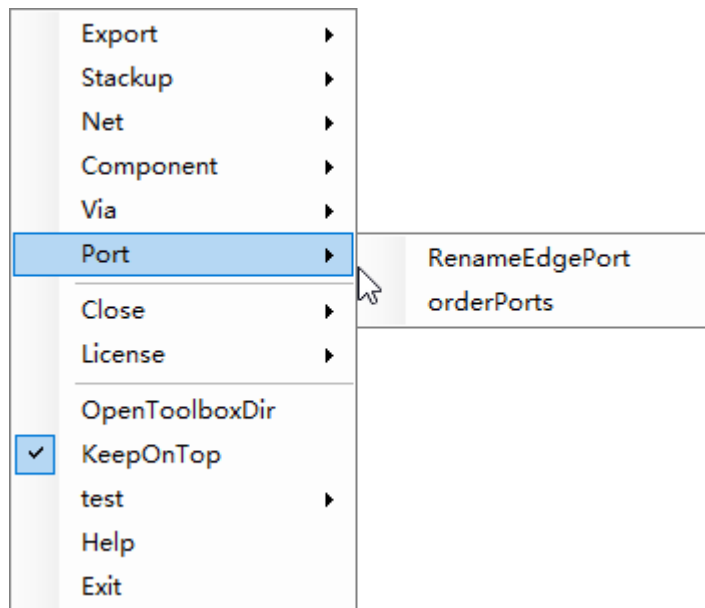
材料名称	规格	单位	数量	单价	合价	材料名称	规格	单位	数量	单价	合价	材料名称	规格	单位	数量	单价	合价	材料名称	规格	单位	数量	单价	合价						
钢筋	Φ12	m	100	1.20	120.00	钢筋	Φ14	m	100	1.40	140.00	钢筋	Φ16	m	100	1.60	160.00	钢筋	Φ18	m	100	1.80	180.00	钢筋	Φ20	m	100	2.00	200.00
钢筋	Φ22	m	100	2.20	220.00	钢筋	Φ24	m	100	2.40	240.00	钢筋	Φ26	m	100	2.60	260.00	钢筋	Φ28	m	100	2.80	280.00	钢筋	Φ30	m	100	3.00	300.00
钢筋	Φ32	m	100	3.20	320.00	钢筋	Φ34	m	100	3.40	340.00	钢筋	Φ36	m	100	3.60	360.00	钢筋	Φ38	m	100	3.80	380.00	钢筋	Φ40	m	100	4.00	400.00
钢筋	Φ42	m	100	4.20	420.00	钢筋	Φ44	m	100	4.40	440.00	钢筋	Φ46	m	100	4.60	460.00	钢筋	Φ48	m	100	4.80	480.00	钢筋	Φ50	m	100	5.00	500.00
钢筋	Φ52	m	100	5.20	520.00	钢筋	Φ54	m	100	5.40	540.00	钢筋	Φ56	m	100	5.60	560.00	钢筋	Φ58	m	100	5.80	580.00	钢筋	Φ60	m	100	6.00	600.00
钢筋	Φ62	m	100	6.20	620.00	钢筋	Φ64	m	100	6.40	640.00	钢筋	Φ66	m	100	6.60	660.00	钢筋	Φ68	m	100	6.80	680.00	钢筋	Φ70	m	100	7.00	700.00
钢筋	Φ72	m	100	7.20	720.00	钢筋	Φ74	m	100	7.40	740.00	钢筋	Φ76	m	100	7.60	760.00	钢筋	Φ78	m	100	7.80	780.00	钢筋	Φ80	m	100	8.00	800.00
钢筋	Φ82	m	100	8.20	820.00	钢筋	Φ84	m	100	8.40	840.00	钢筋	Φ86	m	100	8.60	860.00	钢筋	Φ88	m	100	8.80	880.00	钢筋	Φ90	m	100	9.00	900.00
钢筋	Φ92	m	100	9.20	920.00	钢筋	Φ94	m	100	9.40	940.00	钢筋	Φ96	m	100	9.60	960.00	钢筋	Φ98	m	100	9.80	980.00	钢筋	Φ100	m	100	10.00	1000.00
钢筋	Φ102	m	100	10.20	1020.00	钢筋	Φ104	m	100	10.40	1040.00	钢筋	Φ106	m	100	10.60	1060.00	钢筋	Φ108	m	100	10.80	1080.00	钢筋	Φ110	m	100	11.00	1100.00
钢筋	Φ112	m	100	11.20	1120.00	钢筋	Φ114	m	100	11.40	1140.00	钢筋	Φ116	m	100	11.60	1160.00	钢筋	Φ118	m	100	11.80	1180.00	钢筋	Φ120	m	100	12.00	1200.00
钢筋	Φ122	m	100	12.20	1220.00	钢筋	Φ124	m	100	12.40	1240.00	钢筋	Φ126	m	100	12.60	1260.00	钢筋	Φ128	m	100	12.80	1280.00	钢筋	Φ130	m	100	13.00	1300.00
钢筋	Φ132	m	100	13.20	1320.00	钢筋	Φ134	m	100	13.40	1340.00	钢筋	Φ136	m	100	13.60	1360.00	钢筋	Φ138	m	100	13.80	1380.00	钢筋	Φ140	m	100	14.00	1400.00
钢筋	Φ142	m	100	14.20	1420.00	钢筋	Φ144	m	100	14.40	1440.00	钢筋	Φ146	m	100	14.60	1460.00	钢筋	Φ148	m	100	14.80	1480.00	钢筋	Φ150	m	100	15.00	1500.00
钢筋	Φ152	m	100	15.20	1520.00	钢筋	Φ154	m	100	15.40	1540.00	钢筋	Φ156	m	100	15.60	1560.00	钢筋	Φ158	m	100	15.80	1580.00	钢筋	Φ160	m	100	16.00	1600.00
钢筋	Φ162	m	100	16.20	1620.00	钢筋	Φ164	m	100	16.40	1640.00	钢筋	Φ166	m	100	16.60	1660.00	钢筋	Φ168	m	100	16.80	1680.00	钢筋	Φ17				

UNNAMED_000	TOP
UNNAMED_002	
UNNAMED_004	L2_SIG
UNNAMED_006	L3_GND
UNNAMED_008	L4_SIG
UNNAMED_010	L5_GND
UNNAMED_012	L6_SIG
UNNAMED_014	L7_GND
UNNAMED_016	L8_SIG
UNNAMED_018	L9_GND
UNNAMED_020	L10_SIG
UNNAMED_022	L11_GND
UNNAMED_024	L12_SIG
UNNAMED_026	L13_GND
UNNAMED_028	L14_POW
UNNAMED_030	L15_POW
UNNAMED_032	L16_GND
UNNAMED_034	L17_POW
UNNAMED_036	L18_GND
UNNAMED_038	L19_SIG
UNNAMED_040	L20_GND
UNNAMED_042	L21_SIG
UNNAMED_044	L22_GND
UNNAMED_046	L24_SIG
UNNAMED_048	L24_GND
UNNAMED_050	L25_SIG
UNNAMED_052	L26_GND
UNNAMED_054	L27_SIG
UNNAMED_056	L28_GND
UNNAMED_058	L29_SIG
UNNAMED_060	BOTTOM

Automatically name Port names

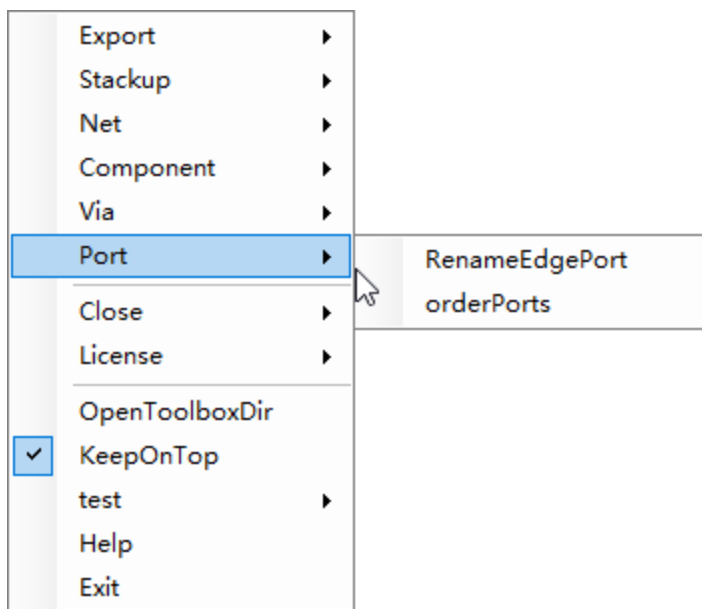
Rename the manually added Ports in the 3D Layout for easier identification.

- If the Port is located on a Pin, name it in the format of refdes.pin.Net, for example, U1.A2.DQ0.
- If the Port is not located on a Pin, name it with the net name.

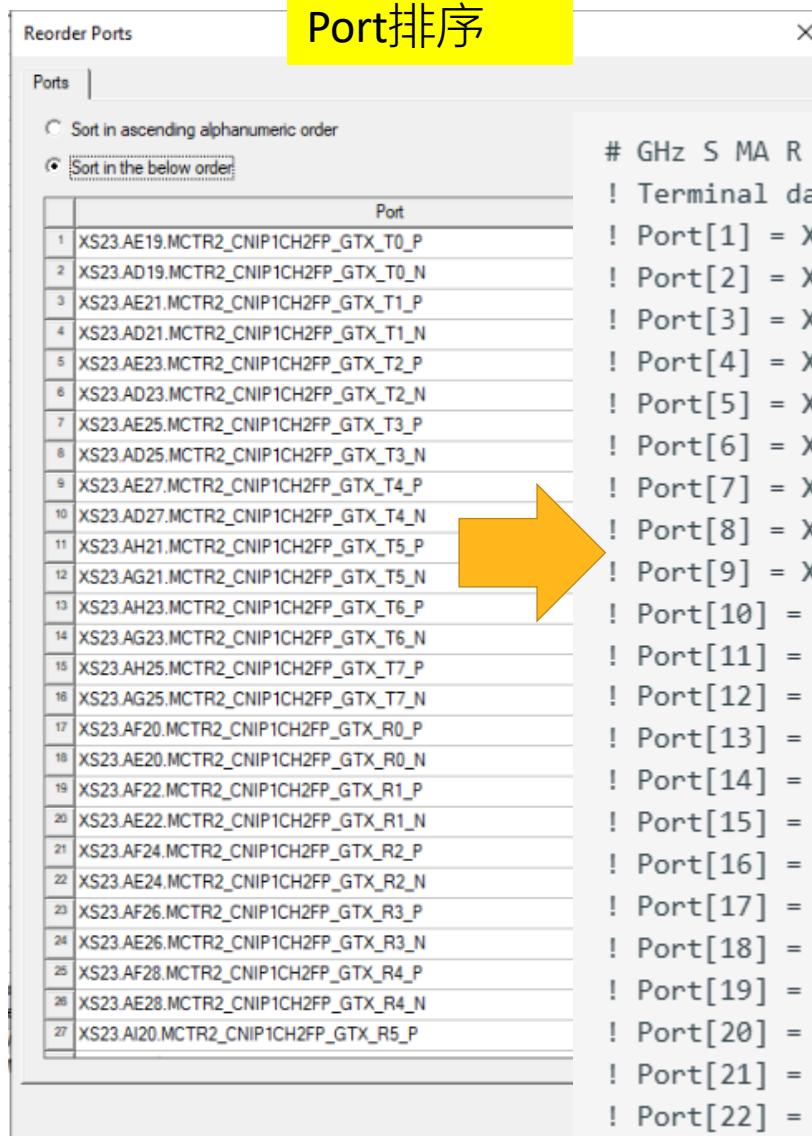


Port sorting

- Sort the Ports in the 3D Layout according to Net, with the following sorting rules: 1. Component Name 2. Bus Number 3. Differential P/N, +/-, P/M



Port排序

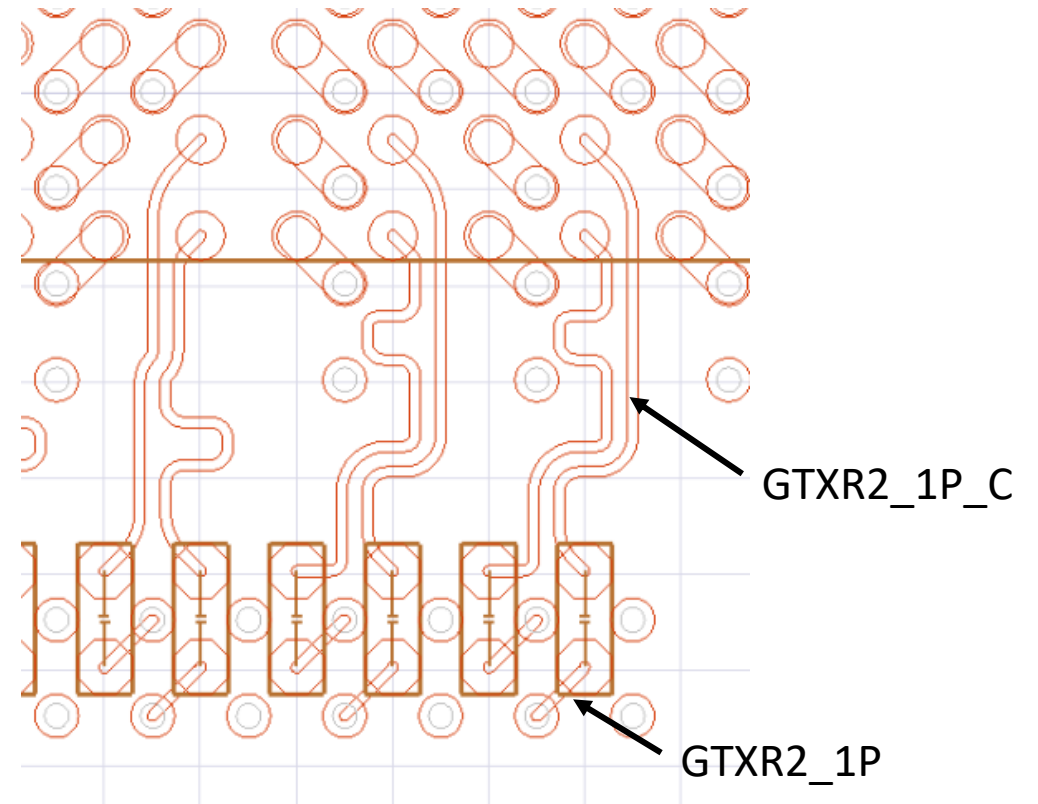
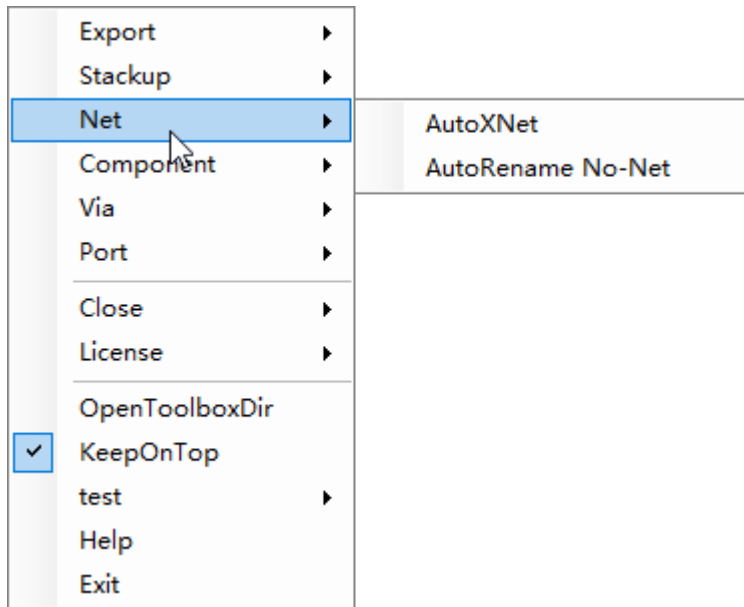


输出S参数

```
# GHz S MA R 50.000000
! Terminal data exported
! Port[1] = XS23.AE19.MCTR2_CNIP1CH2FP_GTX_T0_P
! Port[2] = XS23.AD19.MCTR2_CNIP1CH2FP_GTX_T0_N
! Port[3] = XS23.AE21.MCTR2_CNIP1CH2FP_GTX_T1_P
! Port[4] = XS23.AD21.MCTR2_CNIP1CH2FP_GTX_T1_N
! Port[5] = XS23.AE23.MCTR2_CNIP1CH2FP_GTX_T2_P
! Port[6] = XS23.AD23.MCTR2_CNIP1CH2FP_GTX_T2_N
! Port[7] = XS23.AE25.MCTR2_CNIP1CH2FP_GTX_T3_P
! Port[8] = XS23.AD25.MCTR2_CNIP1CH2FP_GTX_T3_N
! Port[9] = XS23.AE27.MCTR2_CNIP1CH2FP_GTX_T4_P
! Port[10] = XS23.AD27.MCTR2_CNIP1CH2FP_GTX_T4_N
! Port[11] = XS23.AH21.MCTR2_CNIP1CH2FP_GTX_T5_P
! Port[12] = XS23.AG21.MCTR2_CNIP1CH2FP_GTX_T5_N
! Port[13] = XS23.AH23.MCTR2_CNIP1CH2FP_GTX_T6_P
! Port[14] = XS23.AG23.MCTR2_CNIP1CH2FP_GTX_T6_N
! Port[15] = XS23.AH25.MCTR2_CNIP1CH2FP_GTX_T7_P
! Port[16] = XS23.AG25.MCTR2_CNIP1CH2FP_GTX_T7_N
! Port[17] = XS23.AF20.MCTR2_CNIP1CH2FP_GTX_R0_P
! Port[18] = XS23.AE20.MCTR2_CNIP1CH2FP_GTX_R0_N
! Port[19] = XS23.AF22.MCTR2_CNIP1CH2FP_GTX_R1_P
! Port[20] = XS23.AE22.MCTR2_CNIP1CH2FP_GTX_R1_N
! Port[21] = XS23.AF24.MCTR2_CNIP1CH2FP_GTX_R2_P
! Port[22] = XS23.AE24.MCTR2_CNIP1CH2FP_GTX_R2_N
! Port[23] = XS23.AF26.MCTR2_CNIP1CH2FP_GTX_R3_P
```

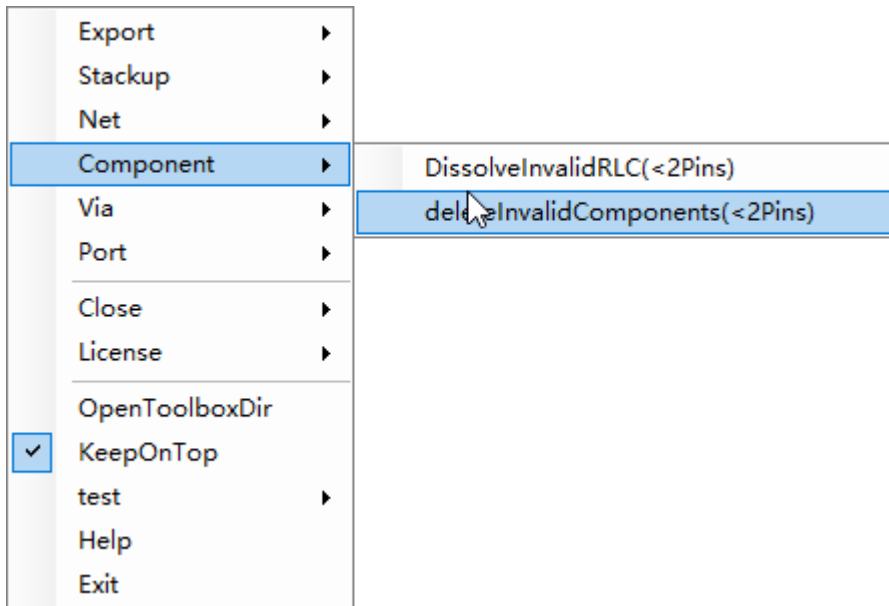
XNets Renaming

- AutoXnet can assist in naming the net after it passes through a capacitor, based on the side that already has a name. The newly named net will be appended with a "_C", "_R", or "_L" suffix depending on the RLC components.



Remove invalid RLC components

- Removing invalid RLC components makes it easier to observe and adjust the configuration of capacitors during PDN analysis.



Active	Part Number	RefDes	Capacitance (F)	Parasitic L (H)	Parasitic R (o...	Positive "Terminal" Net	Negative "Terminal" ...
YES	GRM033R60J104KE19D	C289	1E-07	0	0		N/A
YES	GRM033R60J104KE19D	C290	1E-07	0	0		N/A
YES	GRM155R71H103KA88D	C373	1E-08	0	0		N/A
YES	GRM155R71H103KA88D	C374	1E-08	0	0		N/A
YES	GRM155R71H103KA88D	C375	1E-08	0	0		N/A
YES	GRM155R71H103KA88D	C376	1E-08	0	0		N/A
YES	GRM155R71H103KA88D	C377	1E-08	0	0		N/A
YES	GRM155R71H103KA88D	C378	1E-08	0	0		N/A
YES	GRM155R71H103KA88D	C379	1E-08	0	0		N/A
YES	GRM155R71H103KA88D	C380	1E-08	0	0		N/A
YES	C3216X5R1A476M160AB	C5	4.7E-05	0	0	1V0	GND
YES	C3216X5R1A476M160AB	C6	4.7E-05	0	0	1V0	GND
YES	C3216X5R1A476M160AB	C7	4.7E-05	0	0	1V0	GND
YES	GRM31CR60J107ME39L	C59	0.0001	0	0		
YES	GRM31CR60J107ME39L	C61	0.0001	0	0		
YES	GRM31CR60J107ME39L	C62	0.0001	0	0		
YES	GRM31CR60J107ME39L	C63	0.0001	0	0		
YES	GRM31CR60J107ME39L	C64	0.0001	0	0	1V0	GND
YES	GRM31CR60J107ME39L	C65	0.0001	0	0	1V0	GND
YES	GRM31CR60J107ME39L	C66	0.0001	0	0	1V0	GND

380 capacitors, 380 active, 0 inactive

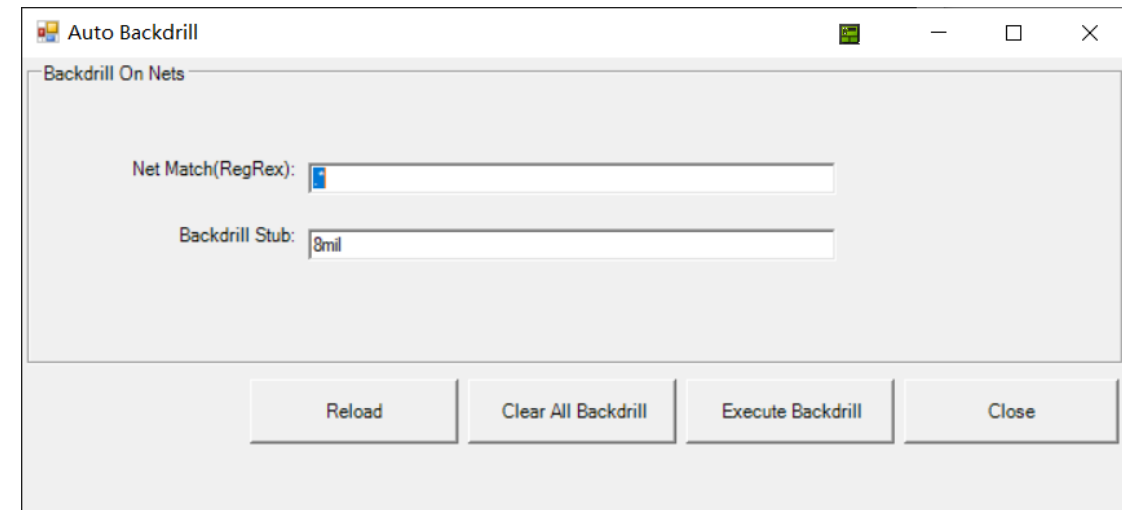
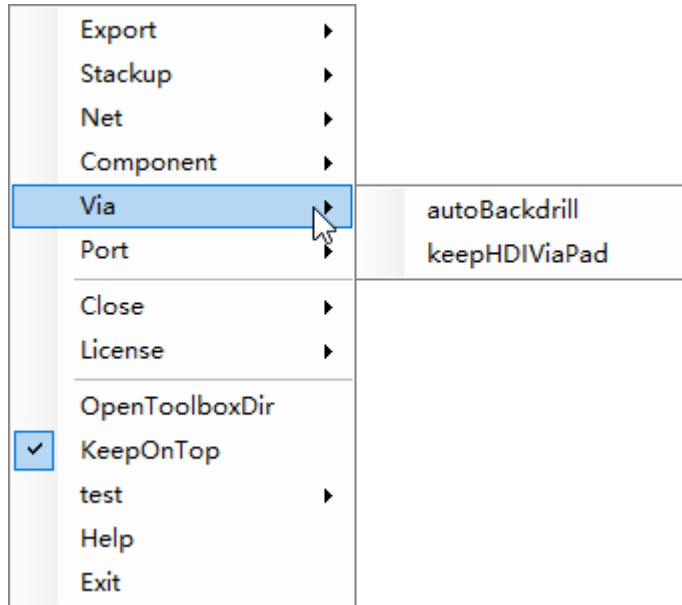
Active	RefDes	Capacitance (F)	Parasitic L (H)	Parasitic R (ohms)	Positive
YES	C378	1E-08	0	0	

Caps will be removed
which pin count <2

Back drilling settings for vias

Set up back drilling for vias according to the given Nets(regex) and Stub information.

- Automatically determine the direction of back drilling.
- (Currently only supported in versions after 2023R1)

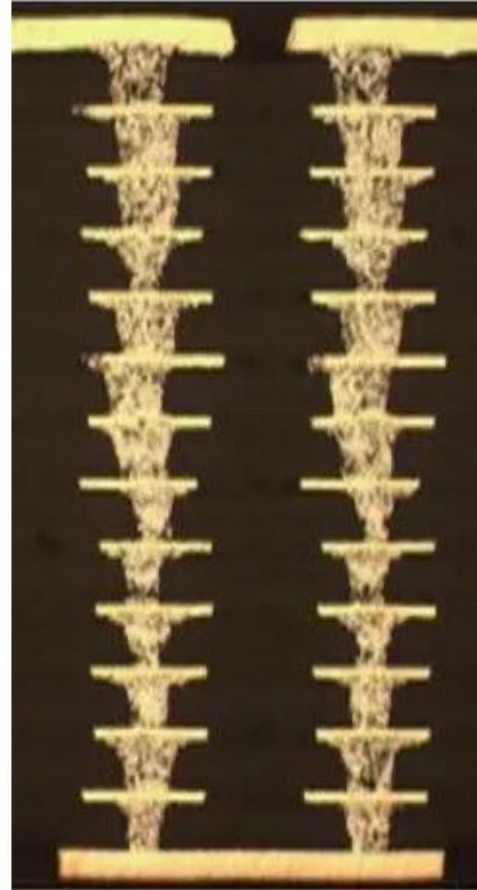
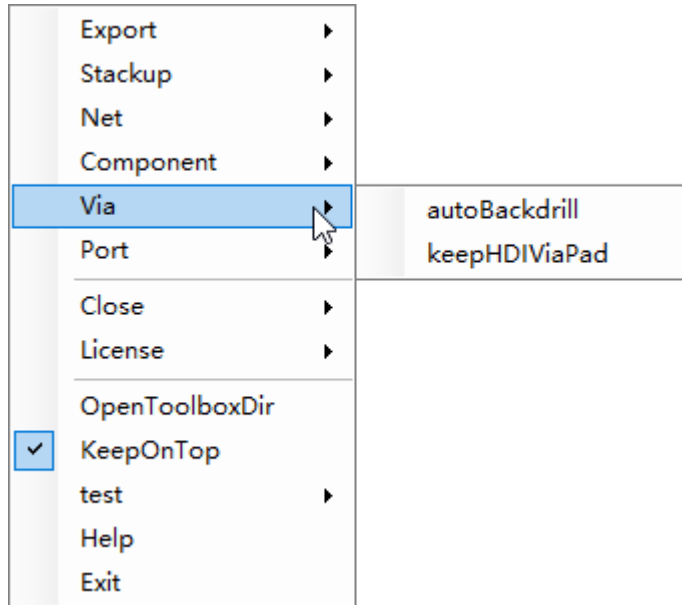


A screenshot of the 'Properties' dialog box for a via. It contains a table with the following data:

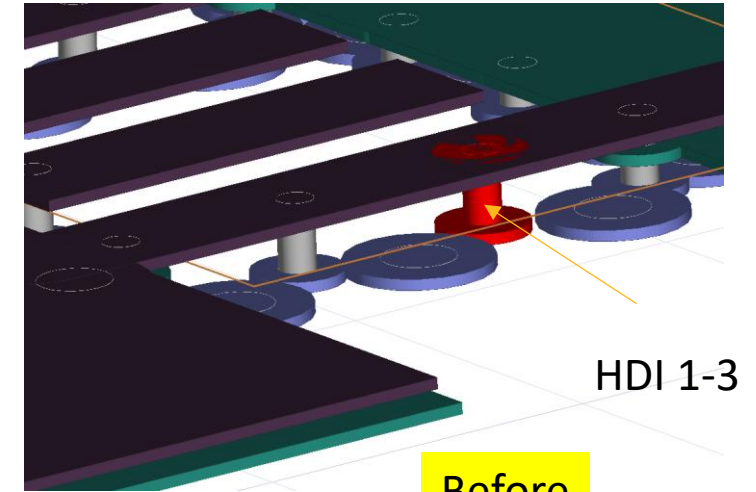
Name	Value	Unit	Evaluated Value
Type	Via		
Lock Position	☐		
Name	Via1856		
Net	GTX_R2_P		
Padstack Definition	v40h25m0...		
Padstack Usage	...		
Start Layer	Top Layer 1		
Stop Layer	Bottom Layer		
Backdrill Top	---		
Top Offset	0	mm	0mm
Backdrill Bottom	ART9		
Bottom Offset	8	mil	8mil
Backdrill Bottom Diameter	9.8425	mil	9.8425mil
Override Hole Diameter	☐		
Hole Diameter	9.8425	mil	9.8425mil
Location	56.1615971...	mm	56.161597mm , ...
Angle	0	deg	0deg

Keep the intermediate pads of Stacked HDI Vias

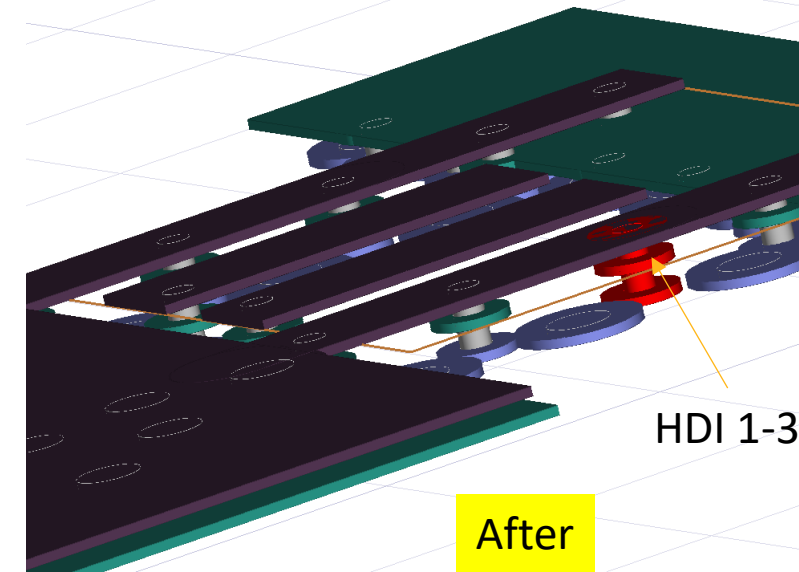
- The KeepHDIViaPad script can detect HDI vias and add circular shapes on each layer to keep these actually existing pads.



Stack HDI Via pads



Before



After

Add your own script

```
menu.xml
1 <?xml version="1.0" encoding="utf-8"?>
2 <Menu>
3   <SubMenu Name="Export" DesignType="3DLayout">
4     <SubMenu Type="MenuItem" Name="ExportToHfssWithNets" ExecuteType="Python" Path="$UserLib/3DLayout/export/export2HfssWithNets.py" Arguments="" PythonPath="" entryFunction="main"></SubMenu>
5   </SubMenu>
6   <SubMenu Name="Stackup" DesignType="3DLayout">
7     <SubMenu Type="MenuItem" Name="ImportCSVStackup" ExecuteType="Python" Path="$UserLib/3DLayout/Stackup/importCSVStackup.py" Arguments="" PythonPath="" entryFunction="main"></SubMenu>
8     <SubMenu Type="MenuItem" Name="ExportCSVStackup" ExecuteType="Python" Path="$UserLib/3DLayout/Stackup/exportCSVStackup.py" Arguments="" PythonPath="" entryFunction="main"></SubMenu>
9   </SubMenu>
10  <SubMenu Name="Net" DesignType="3DLayout">
11    <SubMenu Type="MenuItem" Name="AutoXNet" ExecuteType="Python" Path="$UserLib/3DLayout/Net/autoRLCNet.py" Arguments="" PythonPath=""></SubMenu>
12  </SubMenu>
13  <SubMenu Name="Component" DesignType="3DLayout">
14    <SubMenu Type="MenuItem" Name="deleteInvalidRLC" ExecuteType="Python" Path="$UserLib/3DLayout/component/deleteInvalidRLC.py" Arguments="" PythonPath=""></SubMenu>
15  </SubMenu>
16  <SubMenu Name="Via" DesignType="3DLayout">
17    <SubMenu Type="MenuItem" Name="autoBackdrill" ExecuteType="IronPython" Path="$UserLib/3DLayout/Via/autoBackdrill.py" Arguments="" PythonPath=""></SubMenu>
18  </SubMenu>
19  <SubMenu Name="Port" DesignType="3DLayout">
20    <SubMenu Type="MenuItem" Name="RenameEdgePort" ExecuteType="Python" Path="$UserLib/3DLayout/Port/nameEdgePort.py" Arguments="" PythonPath=""></SubMenu>
21  </SubMenu>
22  <SubMenu Type="Separator"/>
23  <SubMenu Name="Close" DesignType="3DLayout">
24    <SubMenu Type="MenuItem" Name="CloseAllProjectWithoutSave" ExecuteType="Python" Path="$UserLib/Desktop/close.py" EntryFunc="CloseAllProjectWithoutSave"></SubMenu>
25    <SubMenu Type="MenuItem" Name="ReloadProject" ExecuteType="IronPython" Path="$UserLib/Desktop/Close.py" EntryFunc="Reload"></SubMenu>
26  </SubMenu>
27  <SubMenu Name="License" DesignType="3DLayout">
28    <SubMenu Type="MenuItem" Name="licenseConnectionReset(VPN)" ExecuteType="EXE" Path="$UserLib/App/licenseConnectionReset.bat"></SubMenu>
29  </SubMenu>
30 </Menu>
```

By using the menu.xml file in the directory, you can control and add menu content, display methods, script execution paths, etc.

Example of XML Content Definition

```
<Menu>
<SubMenu Name="test">
<SubMenu Type="MenuItem" Name="showProjectName_python" ExecuteType="Python" Path="$UserLib/Template/showProjectName.py"
Arguments ="" PythonPath=""></SubMenu> <SubMenu Type="MenuItem" Name="showProjectName_Ironpython" ExecuteType="IronPython"
Path="$UserLib/Template/showProjectName.py" Arguments ="" EntryFunc="main" PythonPath=""></SubMenu>
</SubMenu>
</Menu>
```

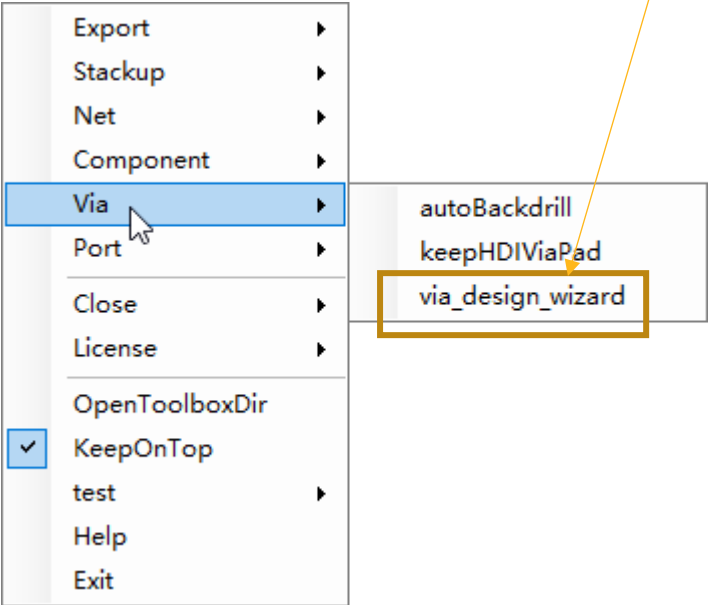
- Type: The value can be "MenuItem" (menu item), "Separator" (separator). If omitted, it defaults to "MenuItem".
- Name: The name displayed on the menu.
- ExecuteType: The type of executable program. Optional values: Python, IronPython, EXE (including bat, openable documents, etc.).
- Path: The path to the executable program or script. Absolute paths can be used. If a relative path is used, it can be introduced in the form of "\$+ folders " to reference folders within the current directory.
- Arguments: Allows passing arguments to the executable script, with multiple arguments separated by spaces.
- PythonPath: You can specify the execution path for Python, especially useful for distinguishing between versions when multiple versions are installed. If omitted, the Python executable will be searched for in the PATH variable.
- EntryFunc: For Python and IronPython, specifies the entry function for running the script. The default is the "main" function (which can be omitted). If it is another function, the function name needs to be specified.

Add toolkit to Menu

Add an existing script. As shown in the figure, add the existing tool via_design_wizard to the plugin.

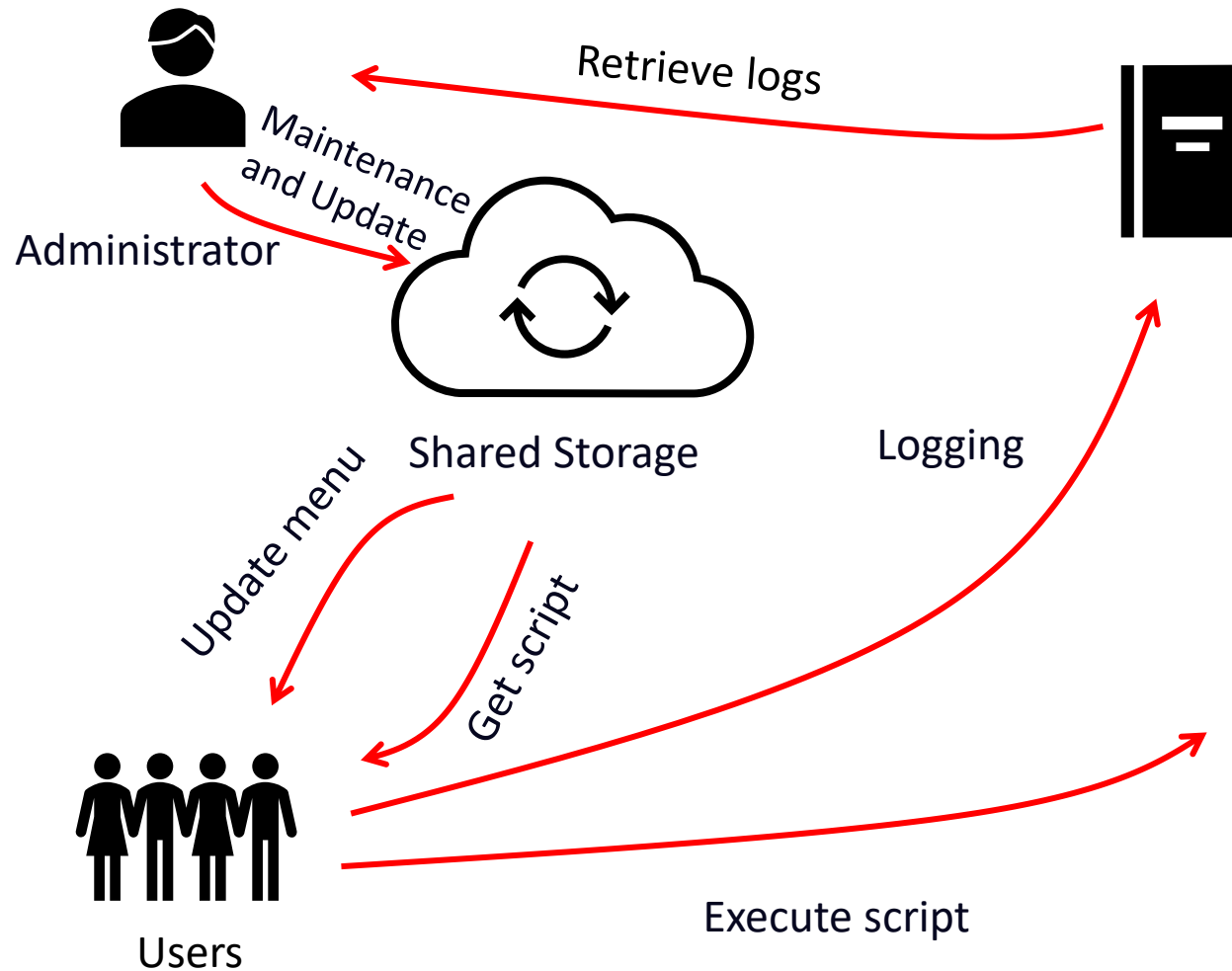
- If the script is placed in the tool directory, a relative path can be used.
- If the script's entry point is the main() function, EntryFunc can be omitted.

```
2 <SubMenu Name="Via" DesignType="3DLayout">
3   <SubMenu Type="MenuItem" Name="autoBackdrill" ExecuteType="IronPython" Path=
4     "$UserLib/3DLayout/Via/autoBackdrill.py" Arguments="" PythonPath=""></SubMenu>
5   <SubMenu Type="MenuItem" Name="keepHDIviaPad" ExecuteType="Python" Path=
6     "$UserLib/3DLayout/Via/keepHDIviaPad.py" Arguments="" PythonPath=""></SubMenu>
7   <SubMenu Type="MenuItem" Name="via_design_wizard" ExecuteType="Python" Path=
8     "C:/work/Project/AE/Script/via_design_wizard-0.0.3/src/via_design_wizard.py"
9     Arguments="" PythonPath=""></SubMenu>
10 </SubMenu>
```

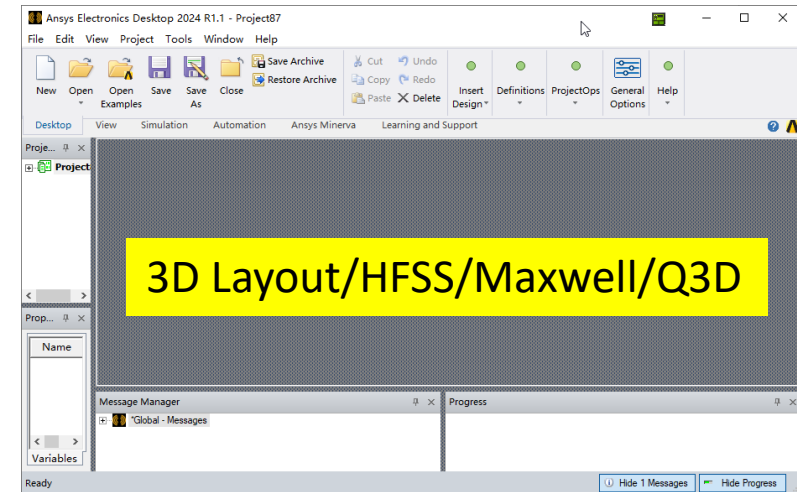


```
1302 def main():
1303     port = get_port()
1304     version = get_aedt_version()
1305     aedt_process_id = get_process_id()
1306
1307     app = ViaDesignFrontend()
1308     app.mainloop()
1309     file_path = app.execute.get("file_path")
1310     if file_path:
1311         vd = ViaDesignConfig(file_path, version)
1312
1313         dt = ansys.aedt.core.Desktop(
1314             new_desktop_session=False,
1315             specified_version=version,
1316             port=port,
1317             aedt_process_id=aedt_process_id,
1318         )
1319         h3d = Hfss3dLayout(vd.edb_path, version)
1320         dt.release_desktop(False, False)
1321
1322 if __name__ == "__main__":
1323     # Open UI
1324     main()
1325
```

Share scripts at the company level



- Automatic recognition of script changes and updates
- Automatic recognition of AEDT windows and versions
- Unified maintenance and logging
- Access control and usage identification



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Note: Pyaedt needs to be installed locally in order to use.

Note: If an offline version is required, please contact Ansys AE.